

Order Statistics:

In practice, the random variables of interest may depend on the relative magnitudes of the observed variable. For example, we may be interested in the maximum mileage per gallon of a particular class of cars.

Definition: let $X_1, X_2, \dots, X_n$ be a random sample from a continuous distribution with pdf $f(x)$. Let $Y_1, Y_2, \dots, Y_n$ be a permutation of $X_1, X_2, \dots, X_n$ such that

$$Y_1 \leq Y_2 \leq \dots \leq Y_n$$

Then the ordered random variables $Y_1, Y_2, \dots, Y_n$ are called the **ordered statistics** of the random sample $X_1, X_2, \dots, X_n$. Here Y_k is called the k^{th} order statistic. Because of continuity, the equality sign could be ignored.

Remark: Although X_i 's are i.i.d. random variables, the random variables Y_i 's are neither independent nor identically distributed.

Thus, the minimum of X_i 's is $Y_1 = \min(X_1, X_2, \dots, X_n)$

and the maximum of X_i 's is $Y_n = \max(X_1, X_2, \dots, X_n)$.

The order statistics of the sample $X_1, X_2, \dots, X_n$ can also be denoted by $X_{(1)}, X_{(2)}, \dots, X_{(n)}$, where,

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

Example:

- (i) The range $R = Y_n - Y_1$ is a function of order statistics.
- (ii) The sample median M equals Y_{m+1} if $n = 2m + 1$.

Hence, the sample median M is an order statistic, when n is odd. If n is even, then the sample median can be obtained using the order statistic,

$$M = \frac{1}{2} \left[Y_{\frac{n}{2}} + Y_{\frac{n}{2}+1} \right]$$

The following result is useful in determining the distribution of functions of more than one order statistics.

Theorem: Let $X_1, X_2, \dots, X_n$ be a random sample from a population with pdf $f(x)$. Then the joint pdf of order statistics $Y_1, Y_2, \dots, Y_n$ is

$$f(y_1, y_2, \dots, y_n) = \begin{cases} n! f(y_1) f(y_2) \dots f(y_n) & \text{for } y_1 < y_2 < \dots < y_n \\ 0 & \text{otherwise} \end{cases}$$

The pdf of the k^{th} order statistic is given by the following theorem.

Theorem: The pdf of Y_k is,

$$f_k(y) = f_{Y_k}(y) = \frac{n!}{(k-1)!(n-k)!} f(y) \{F(y)\}^{k-1} \{1-F(y)\}^{n-k}, \quad \text{for } -\infty < y < \infty$$

where, $F(y) = P(X_i \leq y)$ is the CDF of X_i .

In particular, the pdf of Y_1 is $f_1(y) = nf(y)[1-F(y)]^{n-1}$ and the pdf of Y_n is

$f_n(y) = nf(y)[F(y)]^{n-1}$. In the following example, we will derive pdf of Y_n .

Example 1: Let $X_1, X_2, \dots, X_n$ be a random sample from $U[0,1]$. Find the pdf of the k^{th} order statistic Y_k .

Solution: Since the pdf of X_i is $f(x) = 1, 0 \leq x \leq 1$, the CDF is $F(x) = x, 0 \leq x \leq 1$. The pdf of the k^{th} order statistic Y_k reduces to,

$$f_k(y) = \frac{n!}{(k-1)!(n-k)!} y^{k-1} (1-y)^{n-k}, \quad 0 \leq y \leq 1$$

Which is a Beta distribution with $\alpha = k$ and $\beta = n - k + 1$.

Example 2: Find the distribution of the n^{th} order statistic Y_n of the sample $X_1, X_2, \dots, X_n$ from a population with pdf $f(x)$.

Solution: Let the cdf of Y_n be denoted by $F_n(y)$. Then,

$$\begin{aligned} F_n(y) &= P(Y_n \leq y) = P(\max_{1 \leq i \leq n} X_i \leq y) \\ &= P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y) \\ &= \{F(y)\}^n \quad (\text{by independence}) \end{aligned}$$

Hence, the pdf $f_n(y)$ of Y_n is,

$$\begin{aligned} f_n(y) &= \frac{d}{dy} [F(y)]^n = n[F(y)]^{n-1} \frac{d}{dy} F(y) \\ &= n[F(y)]^{n-1} f(y) \end{aligned}$$

In particular, if $X_1, X_2, \dots, X_n$ is a random sample from $U[0,1]$, then the cumulative extreme value distribution is given by,

$$F_n(y) = \begin{cases} 0, & y < 0 \\ y^n, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

Example 3: A string of 10 light bulbs is connected in series, which means that the entire string will not light up if any one of the light bulbs fails. Assume that the lifetimes of the

bulbs, $\tau_1, \tau_2, \dots, \tau_{10}$, are independent random variables that are exponentially distributed with mean 2. Find the distribution of the life length of this string of light bulbs.

Solution: Note that the pdf of τ_i is $f(t) = 2e^{-2t}$, $0 < t < \infty$, and the cumulative distribution of τ_i is $F_{\tau_i}(t) = 1 - e^{-2t}$, $t > 0$.

Let T represent the lifetime of this string of light bulbs. Then, $T = \min(\tau_1, \tau_2, \dots, \tau_n)$. Thus,

$$F_T(t) = 1 - [1 - F_{\tau_i}(t)]^{10}$$

Hence, the density of T is obtained by differentiating $F_T(t)$ with respect to t , that is,

$$\begin{aligned} f_T(t) &= 10 f_{\tau_i}(t) [1 - F_{\tau_i}(t)]^9 \\ &= \begin{cases} 2(10)e^{-2t} (e^{-2t})^9, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} 20e^{-20t}, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

The joint pdf of the order statistics is given by the following result.

Theorem: Let $X_1, X_2, \dots, X_n$ be a random sample with continuous probability density function $f(x)$ and a distribution function $F(x)$. Let, $Y_1, Y_2, \dots, Y_n$ be the order statistics. Then for any $1 \leq i < k \leq n$ and $-\infty < x \leq y < \infty$, the joint pdf of Y_i and Y_k is given by,

$$f_{Y_i, Y_k}(x, y) = \frac{n!}{(i-1)!(k-i-1)!(n-k)!} [F(x)]^{i-1} [F(y) - F(x)]^{k-i-1} [1 - F(y)]^{n-k} f(x) f(y)$$

Example 4: Let $X_1, X_2, \dots, X_n$ be a random sample from $U[0,1]$. Find the joint pdf of Y_2 and Y_5 .

Solution: Taking $i = 2$ and $k = 5$ in the above theorem, we get the joint pdf of Y_2 and Y_5 as,

$$\begin{aligned} f_{Y_2, Y_5}(x, y) &= \frac{n!}{(2-1)!(5-2-1)!(n-5)!} [F(x)]^{2-1} [F(y) - F(x)]^{5-2-1} [1 - F(y)]^{n-5} f(x) f(y) \\ &= \begin{cases} \frac{n!}{2(n-5)!} x(y-x)^2 (1-y)^{n-5}, & 0 < x \leq y < 1 \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$